Staggered kite finite elements with mass lumping for the shallow-water equations and the compressible Euler equations on polygonal meshes

Oswald Knoth

Abstract

Every polygonal mesh of the sphere or the plane — triangles, quadrilaterals and the hexagons and pentagons of Voronoi meshes — is split into quadrilaterals (*kites*) by joining the cell centres with points on the edges. On the kite mesh we define a height space of tensor Gauss–Radau elements, continuous inside every primal cell and broken across primal edges, and a velocity space of Raviart–Thomas elements, normal-continuous inside every dual cell and broken across its boundary. The height mass matrix is diagonal, the velocity mass matrix block diagonal with one block per dual cell, and the discrete divergence and gradient are exact adjoints and free of the metric. The rotating shallow-water scheme conserves mass and energy exactly and converges at second order on all mesh types. For the pair $\mathrm{RT}_k \times Q_{k+1}$ we prove that the velocity space splits exactly into discrete gradients and curls, at the price of one stationary height mode per primal cell. The extension to the compressible Euler equations is a tensor product with a vertical $\mathrm{DG}_0 \times \mathrm{CG}_1$ discretization that is derived as a Galerkin method with the trapezoidal rule: the vertical mass is diagonal, every Radau node carries an independent column, and the vertical energy exchange is exact. Going from two to three dimensions exposes four problems that the shallow-water tests do not show: a quadrature mismatch between the velocity mass and the projections, a consistency defect of the projected transport of scalars, the instability of the vector-invariant momentum transport in flow faster than the gravity waves — which the higher vertical modes of a stratified atmosphere always are — and the different weighting of the sub-cell height modes by the horizontal and the vertical pressure gradient. A discontinuous Galerkin transport of momentum, density and potential temperature removes the instabilities. The model is quiet on a steady thermal wind, reproduces the linear growth rate

of baroclinic instability in an f-plane channel and gives the same baroclinic wave on hexagonal, cubed-sphere and triangular meshes.

Keywords: staggered finite elements, mass lumping, dual cell method, polygonal meshes, shallow-water equations, compressible Euler equations, HEVI

1. Introduction

Compatible (mimetic) finite elements give the discrete operators of geophysical fluid dynamics the structure of the continuous ones: the gradient is the negative adjoint of the divergence, the curl of a gradient vanishes, and the semi-discrete equations conserve mass and energy [4, 9, 11]. Their extension to three dimensions uses extruded elements whose vertical part is again a compatible pair [11, 10]. The price is that the mass matrices of the usual pairs (e.g. $\mathrm{RT}_k \times \mathrm{DG}_k$) are not diagonal, so an explicit step needs global solves. Finite-volume C grids [13, 12, 22] avoid them with diagonal Hodge stars, at the price of first-order operators on general meshes.

Two families of methods obtain local mass matrices from staggered finite elements. Staggered discontinuous Galerkin (SDG) methods [2, 3, 5] use two staggered macro-element meshes, with one field continuous in the primal and the other in the dual macro elements. Leng, Lan and Ju [8] built from them a conservative SDG method for the shallow-water equations on the sphere: they split every polygon into triangles around an interior point, take the velocity normal-continuous in the polygons and the height continuous in the pairs of triangles around the edges, and prove mass and energy conservation, the absence of spurious pressure modes and a steady geostrophic mode. The dual cell method of Wess, Kapidani, Codecasa and Schöberl [18] splits the primal cells into quadrilaterals around the primal vertices and uses tensor-product spaces whose degrees of freedom are quadrature points, so that the mass matrices are diagonal or block diagonal to arbitrary order; they apply it to acoustic and electromagnetic waves.

This paper combines the two ideas for geophysical flows. The quadrilaterals of the dual cell method, here called *kites*, give tensor-product spaces with Gauss–Radau nodes and a diagonal height mass; the Raviart–Thomas velocity of the SDG methods gives a mass flux with a block-diagonal mass and exactly adjoint, metric-free operators. The heights are continuous in the primal cells and the velocity in the dual cells, the opposite of [8]. The

contributions are:

1. The kite spaces $H_p \times U_k \times Z_{k+1}$ on arbitrary polygonal meshes of the sphere with curved elements, their mass-lumped inner products and the metric-free divergence and gradient (Sections 2–4).
2. For the default pair $U_1 \times H_2$ an exact discrete Helmholtz decomposition of the velocity space and the characterization of its kernel: one stationary height mode per primal cell, given in closed form (Theorem 1).
3. An energy-conserving nonlinear shallow-water scheme with these spaces (Section 5) and its validation on hexagonal, triangular and quadrilateral meshes (Section 6).
4. A vertical discretization of the compressible Euler equations derived as a Galerkin method with the trapezoidal rule, with a diagonal vertical mass, independent columns at the Radau nodes and an exact vertical energy exchange, and its horizontally explicit, vertically implicit (HEVI) time discretization (Section 7).
5. An analysis of what changes from two to three dimensions (Section 8) and a discontinuous Galerkin (DG) transport that removes the resulting instabilities (Section 9), validated on a thermal wind, a baroclinic channel and the baroclinic wave on the sphere (Section 10).

Section 11 compares the method with the related ones; Section 12 concludes.

2. Kite meshes and mass-lumped inner products

2.1. Kite meshes

Let $\mathcal{P}$ be a polygonal primal mesh of a closed surface Ω (the sphere of radius a or a doubly periodic plane) with N cells, E edges and V vertices. Every cell $P \in \mathcal{P}$ with vertices $w_1, \dots, w_n$ is split into n kites

$$K_i = (w_i,\ m_i,\ c_P,\ m_{i-1}), \qquad i = 1, \dots, n,$$

where c_P is the cell centre and m_i a point on the primal edge (w_i, w_{i+1}) (Figure 1). The kites around a primal vertex w form its dual cell $\widetilde{P}_w$, and $\mathcal{K}$ denotes the set of all $2E$ kites. Every kite has the same local numbering: vertex 1 is the primal vertex and vertex 3 the centre. Local edges 1 and 4 are halves of primal edges (*primal faces*); local edges 2 and 3 join an edge point to the centre (*dual faces*) and form the boundaries of the dual cells. Every primal face is interior to a dual cell, every dual face interior to a primal cell.

Cell centres and edge points. The centre is the vertex of the orthogonal dual mesh when there is one: the circumcentre of a triangle, the generator of a Voronoi cell. The generators of a Voronoi mesh are recovered from the cells as the null vector of the homogeneous linear system $g_i - g_j \parallel \mathbf{n}_e$, $\mathbf{n}_e \cdot (g_i + g_j) = 0$ over all edges e between cells i and j; for the published MPAS and ICON meshes this reproduces the stored generators to 10^{-14}. On other meshes (cubed sphere, planes) the centre is the centroid. With an orthogonal dual, m_i is the crossing of the primal edge with the dual edge, so the dual faces lie on the dual edges; otherwise m_i is the edge midpoint.

2.2. Reference map

Every kite is the image of the reference square $\hat{K} = [-1, 1]^2$ with the primal vertex at $(-1, -1)$ and the centre at $(1, 1)$, so that the primal faces are $\{s = -1\}$ and $\{t = -1\}$ and the dual faces $\{s = 1\}$ and $\{t = 1\}$. The map F_K is the bilinear interpolant of the four vertices, on the sphere followed by the radial projection $x \mapsto a\, x/|x|$ (curved elements). $J = DF_K$ is the 3×2 Jacobian and $\det J = |J_1 \times J_2|$ the area element; the outward unit normal of the surface is $\hat{\mathbf{r}} = J_1 \times J_2 / |J_1 \times J_2|$.

2.3. Mass-lumped inner products

Let $r_0 < \cdots < r_p = 1$ be the $p + 1$ Gauss–Radau points of $[-1, 1]$ that include the end point 1 (for $p = 0$ the single point 1), with weights $\omega_0, \ldots, \omega_p$. The tensor rule $\{(r_a, r_b), \omega_a \omega_b\}$ on $\hat{K}$ integrates Q_{2p} exactly. Its nodes lie inside the kite (p^2), on the two dual faces (p each) and at the centre, never on a primal face. The lumped L^2 inner product of two scalar fields is

$$(\eta, \phi)_h = \sum_{K \in \mathcal{K}} \sum_{a,b=0}^{p} \omega_a \omega_b \det J(r_a, r_b)\, \eta(F_K(r_a, r_b))\, \phi(F_K(r_a, r_b)). \qquad (1)$$

On flat kites $\det J$ is bilinear, so for $p \geq 1$ the rule integrates $\det J$ times Q_{2p-1} exactly and the lumped mass of a constant is its exact mass. Velocity inner products and the nonlinear terms use a Gauss rule of $(k + 2)^2$ points per kite with precomputed metric terms.

3. Finite-element spaces

Definition 1 (Height space)**.** H_p is the space of functions that are Q_p on every kite, mapped with F_K, with the Lagrange nodes at the tensor Radau points of Section 2.3:

$$H_p = \{\eta \in L^2(\Omega) : \eta|_P \in H^1(P)\ \forall P \in \mathcal{P},\ \eta|_K \circ F_K \in Q_p\ \forall K \in \mathcal{K}\}.$$

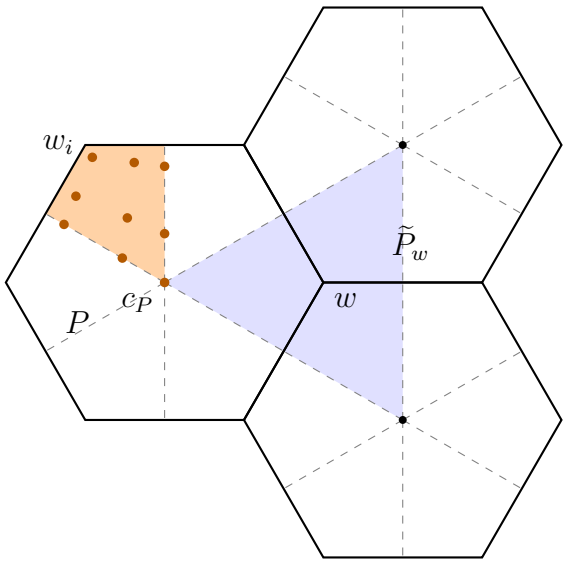


Figure 1: Kite mesh of a hexagonal mesh: primal edges (solid), dual faces from the centres to the edge points (dashed). Shaded blue: the dual cell $\widetilde{P}_w$ of the vertex w, made of the three kites that touch it. Orange: one kite of the cell P with the nine Gauss–Radau nodes of H_2; they lie inside the kite, on its dual faces and at the centre, never on primal edges.

Since no node lies on a primal face, a height field is continuous inside a primal cell and broken across primal edges. With the nodes as quadrature points the mass matrix M_h of (1) is diagonal.

Definition 2 (Velocity space)**.** U_k is the space of fields that are $\mathrm{RTCF}_k = Q_{k+1,k} \times Q_{k,k+1}$ on every kite, mapped with the contravariant Piola transform $\mathbf{v} = J\hat{\mathbf{v}}/\det J$, with normal continuity across the primal faces and none across the dual faces:

$$U_k = \{\mathbf{u} \in L^2(\Omega)^2 : \mathbf{u}|_{\widetilde{P}} \in H(\mathrm{div}, \widetilde{P})\ \forall \widetilde{P} \in \widetilde{\mathcal{P}},\ \hat{\mathbf{u}}_K \in \mathrm{RTCF}_k\ \forall K \in \mathcal{K}\}.$$

The normal degrees of freedom of the dual faces belong to the kite itself, so the velocity mass matrix M_u is block diagonal with one block per dual cell. Two alternatives with the same continuity pattern are implemented: broken BDM_k, and the $Q_p \times Q_p$ space of Wess et al. [18] with Radau nodes at the primal-vertex side, whose lumped mass couples only the two components at one node.

Definition 3 (Vorticity space)**.** Z_{k+1} is the continuous Q_{k+1} space on the kites with Gauss–Lobatto nodes and lumped mass.

Table 1: Degrees of freedom per primal cell and largest mass-matrix block, for the hexagonal ($n = 6$, dual cells of 3 kites), quadrilateral ($n = 4$, 4 kites) and triangular ($n = 3$, about 6 kites) meshes.

	per primal cell			M_u block		
space	hexagons	quads	triangles	hexagons	quads	triangles
H_1	13	9	7	1	1	1
H_2	37	25	19	1	1	1
U_0	18	12	9	9	12	18
U_1	60	40	30	30	40	60
U_2	126	84	63	63	84	126
broken BDM_1	36	24	18	18	24	36
$Q_1 \times Q_1$ of [18]	36	24	18	2	2	2
$Q_2 \times Q_2$ of [18]	90	60	45	2	2	2

The default is $U_1 \times H_2 \times Z_2$. Table 1 lists the dimensions and the sizes of the mass-matrix blocks, Table 2 the six pairs of the experiments. $\mathrm{BDM}_1 \times H_2$ has fewer velocity than height degrees of freedom and is included as a negative example. Per primal cell with n sides,

$$\dim H_p|_P = 1 + n(p + p^2), \qquad \dim U_k|_P = n(k+1)(2k+3),$$

and the M_u block of a dual cell with n_w kites has size $n_w(k+1)(2k+3)$. Globally, with $V - E + N = 2$ on the sphere,

$$\dim H_2 = N + 12E, \qquad \dim U_1 = 20E, \qquad \dim Z_2 = N + V + 7E. \tag{2}$$

4. Discrete operators

With the contravariant Piola transform one has $\mathbf{v} \cdot \nabla\eta \, dA = \hat{\mathbf{v}} \cdot \hat{\nabla}\hat{\eta} \, d\xi$, $(\nabla \cdot \mathbf{v})\, \eta \, dA = (\hat{\nabla} \cdot \hat{\mathbf{v}})\, \hat{\eta} \, d\xi$ and $\mathbf{v} \cdot \mathbf{n} \, ds = \hat{\mathbf{v}} \cdot \hat{\nu} \, d\hat{s}$, so the two operators below are free of the metric and are computed exactly on the reference square.

Definition 4. For $\eta \in H_p$ and $\mathbf{F} \in U_k$ the discrete divergence is

$$D(\eta; \mathbf{F}) = \sum_K \int_K \mathbf{F} \cdot \nabla\eta - \sum_K \sum_{e \subset \partial K \text{ primal}} \int_e \eta_K \, \mathbf{F} \cdot \mathbf{n}_K, \tag{3}$$

Table 2: The pairs of the experiments on a hexagonal mesh: degrees of freedom per primal cell in 2D and in 3D with 30 layers ($30 \dim U + 89 \dim H$ for $\mathbf{u}$, ρ, $\rho\theta$ and w), the ratio of velocity to height degrees of freedom (2 for the continuous fields), and the dimension of $\ker B^T$ (Section 4; N primal cells).

pair	label	$\mathbf{u}$	h	$\dim U / \dim H$	3D, 30 layers	$\dim \ker B^T$
$U_1 \times H_1$	rt1q1	60	13	4.62	2957	1
$U_1 \times H_2$	rt1q2	60	37	1.62	5093	$N+1$
$\mathrm{BDM}_1 \times H_1$	bdm1q1	36	13	2.77	2237	1
$\mathrm{BDM}_1 \times H_2$	bdm1q2	36	37	0.97	4373	$\geq \dim H - \dim U$
$Q_1^2 \times H_1$ [18]	wess1	36	13	2.77	2237	1
$Q_2^2 \times H_2$ [18]	wess2	90	37	2.43	5993	1

with the trace η_K from inside K, and $B_{ij} = D(\eta_i; \mathbf{v}_j)$. For $\pi \in H_p$ and $\mathbf{v} \in U_k$ the discrete gradient is

$$G(\pi; \mathbf{v}) = \sum_K \int_K \pi \, \nabla \cdot \mathbf{v} - \sum_K \sum_{e \subset \partial K \text{ dual}} \int_e \pi \, \mathbf{v}_K \cdot \mathbf{n}_K,$$

where π is single valued on the dual faces.

Proposition 1. *$G(\pi; \mathbf{v}) = -D(\pi; \mathbf{v})$, i.e. $G = -B^T$, and $\mathbf{1}^T B = 0$.*

Proof. On each kite $\int_K \pi \, \nabla \cdot \mathbf{v} = -\int_K \nabla\pi \cdot \mathbf{v} + \int_{\partial K} \pi \, \mathbf{v} \cdot \mathbf{n}_K$. The boundary integral over the dual faces cancels the face term of G, leaving $-\sum_K \int_K \nabla\pi \cdot \mathbf{v} + \sum_K \sum_{\text{primal}} \int \pi_K \, \mathbf{v} \cdot \mathbf{n}_K = -D(\pi; \mathbf{v})$. For $\eta \equiv 1$, $D(1; \mathbf{v}) = -\sum_K \sum_{\text{primal}} \int \mathbf{v} \cdot \mathbf{n}_K = 0$, since $\mathbf{v} \cdot \mathbf{n}$ is continuous across primal faces and both sides appear with opposite normals. □

Vorticity, kinetic energy and Coriolis term. The relative vorticity $\zeta \in Z_{k+1}$ is defined weakly by $(\zeta, \gamma)_h = -\int \mathbf{u} \cdot \mathbf{curl}\gamma$, $\mathbf{curl}\gamma = \hat{\mathbf{r}} \times \nabla\gamma$, the kinetic energy $K \in H_p$ by $(K, \eta)_h = \int \frac{1}{2}|\mathbf{u}|^2 \eta$, and the Coriolis matrix $C_{ij} = \int f \, \mathbf{v}_i \cdot (\hat{\mathbf{r}} \times \mathbf{v}_j)$ is antisymmetric.

4.1. Discrete Helmholtz decomposition and stationary height modes

The curl of Z_{k+1} lies in U_k: on the reference square $\widehat{\mathbf{curl}}\, Q_{k+1} \subset \mathrm{RTCF}_k$, the contravariant Piola transform maps it to the curl on the kite, and the tangential continuity of γ gives normal continuity of $\mathbf{curl}\gamma$ across all faces.

Lemma 1. *$D(\eta; \mathbf{curl}\gamma) = 0$ for all $\eta \in H_p$, $\gamma \in Z_{k+1}$. Hence the discrete gradients $\nabla_h H_p = M_u^{-1} B^T H_p$ and the curls $\mathbf{curl} Z_{k+1}$ are orthogonal in the inner product of M_u.*

Proof. $\mathbf{curl}\gamma$ is divergence free, so $\int_K \mathbf{curl}\gamma \cdot \nabla\eta = \int_{\partial K} \eta_K \, \mathbf{curl}\gamma \cdot \mathbf{n}_K$. The primal faces cancel the face term of (3). On the dual faces both η and $\mathbf{curl}\gamma \cdot \mathbf{n}$ are continuous and appear with opposite normals, so they cancel as well. Finally $(\mathbf{curl}\gamma, M_u^{-1} B^T \eta)_{M_u} = \eta^T B \, \mathbf{curl}\gamma = 0$. □

Theorem 1. *Let $p = k + 1$. For every primal cell P the function ψ_P that equals $L_p(s)L_p(t)$ on every kite of P, in the reference coordinates of that kite, and vanishes outside P, lies in H_p and satisfies $B^T\psi_P = 0$; L_p is the Legendre polynomial of degree p. Consequently*

$$\dim \ker B^T \geq N + 1.$$

Proof. ψ_P is continuous inside P: on the dual face $\{s = 1\}$ of a kite it is $L_p(1)L_p(t) = L_p(t)$, and the neighbouring kite meets this face as $\{t = 1\}$ with the same parameter, so the traces agree. For $\mathbf{v} \in U_k$, with the metric-free form of (3) on each kite of P,

$$\int_{\hat{K}} \Big(\hat{v}_1 \, L_p'(s) L_p(t) + \hat{v}_2 \, L_p(s) L_p'(t) \Big) \, d\xi \\ - L_p(-1) \Big(\int_{-1}^{1} \hat{v}_1(-1, t) L_p(t) \, dt + \int_{-1}^{1} \hat{v}_2(s, -1) L_p(s) \, ds \Big)$$

up to signs of the normals. $\hat{v}_1 \in Q_{k+1,k}$ has degree $k = p - 1$ in t and $\hat{v}_2$ degree $p - 1$ in s, so every integral vanishes by the orthogonality of L_p to polynomials of lower degree. The ψ_P of different cells have disjoint supports, and no combination of them is constant, so they and the constant span a space of dimension $N + 1$ in $\ker B^T$. □

Remark 1. The rank of B was computed for $U_k \times H_{k+1}$, $k = 0, 1, 2$, on hexagonal, cubed-sphere and triangular meshes; in all cases $\dim \ker B^T = N + 1$ exactly. With (2) and $\dim \mathbf{curl} Z_2 = \dim Z_2 - 1$ this gives

$$\dim U_1 = (\dim H_2 - N - 1) + (\dim Z_2 - 1), \qquad U_1 = \nabla_h H_2 \oplus \mathbf{curl} Z_2,$$

an exact discrete Helmholtz decomposition (the sum is orthogonal by Lemma 1, and the dimensions add up with $N + V = E + 2$): no velocity field has both zero discrete divergence and zero discrete vorticity. The pairs with equal order behave the other way round. For $U_1 \times H_1$, $\mathrm{BDM}_1 \times H_1$ and the pair of [18] with $p = 1, 2$, $\ker B^T$ contains only the constant, but for $U_1 \times H_1 \times Z_2$ there are $\dim U_1 - (\dim H_1 - 1) - (\dim Z_2 - 1) = 8E - N$ velocity fields with zero discrete divergence that are orthogonal to all curls.

The modes ψ_P are stationary: for $\mathbf{u} = 0$, $h = \psi_P$ the linear shallow-water tendency below vanishes. In the nonlinear scheme $\psi_P^T M_h \dot{h} = (B^T \psi_P)^T \mathbf{F} = 0$, so the N coefficients $\psi_P^T M_h h$ are exact invariants, one per cell. For smooth fields they are small, since ψ_P is orthogonal to Q_{p-1} in each kite; the gradient converges at second order (Section 6). Section 8.4 shows why the modes matter in three dimensions.

5. Shallow-water schemes

The rotating shallow-water equations in vector-invariant form are

$$\partial_t \mathbf{u} + q\,\hat{\mathbf{r}} \times \mathbf{F} + \nabla\Big(K + gh\Big) = 0, \qquad \partial_t h + \nabla \cdot \mathbf{F} = 0, \tag{4}$$

with the mass flux $\mathbf{F} = h\mathbf{u}$, $K = \frac{1}{2}|\mathbf{u}|^2$ and the potential vorticity $q = (\zeta + f)/h$. They conserve mass and the energy $E = \int \frac{1}{2} h|\mathbf{u}|^2 + \frac{1}{2} g h^2$.

Linear scheme. $M_h \dot{h} = H\, B\, \mathbf{u}$, $M_u \dot{\mathbf{u}} = -C\mathbf{u} - g\, B^T h$ with the mean depth H conserves mass ($\mathbf{1}^T B = 0$) and the energy $\frac{1}{2} H \mathbf{u}^T M_u \mathbf{u} + \frac{1}{2} g h^T M_h h$, since C is antisymmetric and $G = -B^T$.

Nonlinear scheme. The mass flux is the L^2 projection into the velocity space, $\mathbf{F} = \Pi_U(h\mathbf{u})$, i.e. $M_u \mathbf{F} = (\int h\mathbf{u} \cdot \mathbf{v}_i)_i$, one solve with the block-diagonal M_u, and q is evaluated at the quadrature points:

$$M_h \dot{h} = B\mathbf{F}, \qquad \int \dot{\mathbf{u}} \cdot \mathbf{v} = -\int q\,(\hat{\mathbf{r}} \times \mathbf{F}) \cdot \mathbf{v} - B^T(gh + K). \tag{5}$$

Theorem 2. *The scheme* (5) *conserves the mass* $\mathbf{1}^T M_h h$*, the* N *cell invariants* $\psi_P^T M_h h$ *of Theorem 1 if* $p = k + 1$*, and the discrete energy* $E_h = \int \frac{1}{2} h|\mathbf{u}|^2 + \frac{1}{2} g\, h^T M_h h$*.*

Proof. Mass and cell invariants: $\eta^T M_h \dot{h} = (B^T \eta)^T \mathbf{F} = 0$ for $\eta \in \ker B^T$. Energy: with the same quadrature in E_h and the right-hand sides, $\partial E_h / \partial \mathbf{u} = M_u \mathbf{F}$ and $\partial E_h / \partial h = M_h(K + gh)$, so $\dot{E}_h = -\int q\,(\hat{\mathbf{r}} \times \mathbf{F}) \cdot \mathbf{F} - \mathbf{F}^T B^T (gh + K) + (gh + K)^T B \mathbf{F} = 0$. □

This is the framework of McRae and Cotter [9] on the kite spaces, with mass matrices that stay diagonal or block diagonal. Energy conservation is an algebraic property and holds for every quadrature rule of the nonlinear terms. The time discretization is the strong-stability-preserving RK3 scheme. The terms are KernelAbstractions kernels over the kites with atomic scatter and run unchanged on CPUs and GPUs; the model is distributed with MPI over equal-area partitions of the primal cells, one ring of ghost cells deep.

6. Shallow-water experiments

Operator consistency. L^2 errors of the discrete gradient $M_u^{-1}B^T\Pi p$ and divergence $M_h^{-1}B\,\Pi\mathbf{u}$ against the projections of the exact operators, for smooth p and $\mathbf{u}$, on hexagon kites (960, 3840, 15 360 kites) and cubed-sphere kites (384, 1536, 6144):

pair	mesh	gradient	divergence
$Q_0 \times \mathrm{RT}_0$	hexagons	7.4, 7.6, 7.7	0.017, 0.0069, 0.0030
	cube	5.6, 5.8, 5.9	0.065, 0.076, 0.067
$Q_1 \times \mathrm{RT}_0$	hexagons	0.10, 0.052, 0.026	0.23, 0.21, 0.21
	cube	0.19, 0.094, 0.047	0.40, 0.36, 0.31
$Q_1 \times \mathrm{RT}_1$	hexagons	0.21, 0.11, 0.052	0.016, 0.0060, 0.0027
	cube	0.31, 0.15, 0.077	0.044, 0.022, 0.012
$Q_2 \times \mathrm{RT}_1$	hexagons	0.0038, 0.00091, 0.00023	0.088, 0.045, 0.023
	cube	0.0091, 0.0023, 0.00057	0.059, 0.035, 0.019

At the lowest order one of the two operators is inconsistent: with Q_0 heights the pressure is constant per primal cell and does not control the velocity degrees of freedom of the dual faces, and the divergence of broken RT_0 tested with Q_1 does not converge. From RT_1 on both converge.

Williamson test case 2. The steady geostrophic flow [21] converges at second order on all meshes. Linear model, $Q_1 \times \mathrm{RT}_1$, RK4, $\|h(T) - h(0)\|$:

primal mesh	coarse	medium	fine
hexagons (162, 642, 2562 cells)	$3.09 \cdot 10^{-3}$	$7.72 \cdot 10^{-4}$	$1.94 \cdot 10^{-4}$
triangles (80, 320, 1280)	$9.67 \cdot 10^{-3}$	$2.46 \cdot 10^{-3}$	$6.16 \cdot 10^{-4}$
cubed sphere (96, 384, 1536)	$7.88 \cdot 10^{-3}$	$1.94 \cdot 10^{-3}$	$4.88 \cdot 10^{-4}$

Nonlinear model, $\mathrm{RT}_1 \times Q_2 \times \mathrm{CG}_2$, SSP-RK3: $3.60 \cdot 10^{-4}$ and $8.80 \cdot 10^{-5}$ on hexagons 2 and 3 (rate 2.03), mass and energy conserved to round-off. Flat (chordal) kites give rate 1.87. The velocity pairs compare as follows (linear geostrophic state with Coriolis, finest hexagon mesh, rate in brackets):

pair (velocity / height)	gradient	divergence	geostrophic
RT_1 / Q_1	0.21, 0.11, 0.052	0.016, 0.0060, 0.0027	$1.94 \cdot 10^{-4}$ (1.99)
RT_1 / Q_2	0.0038, 0.00091, 0.00023	0.088, 0.045, 0.023	$4.12 \cdot 10^{-5}$ (2.10)
BDM_1 / Q_1	0.13, 0.065, 0.032	0.28, 0.26, 0.25	$1.46 \cdot 10^{-4}$ (1.78)
Wess et al., $p = 1$	0.14, 0.069, 0.035	0.21, 0.18, 0.18	$1.68 \cdot 10^{-4}$ (1.86)
Wess et al., $p = 2$	0.0070, 0.0018, 0.00044	0.026, 0.013, 0.0064	$5.61 \cdot 10^{-6}$ (2.07)

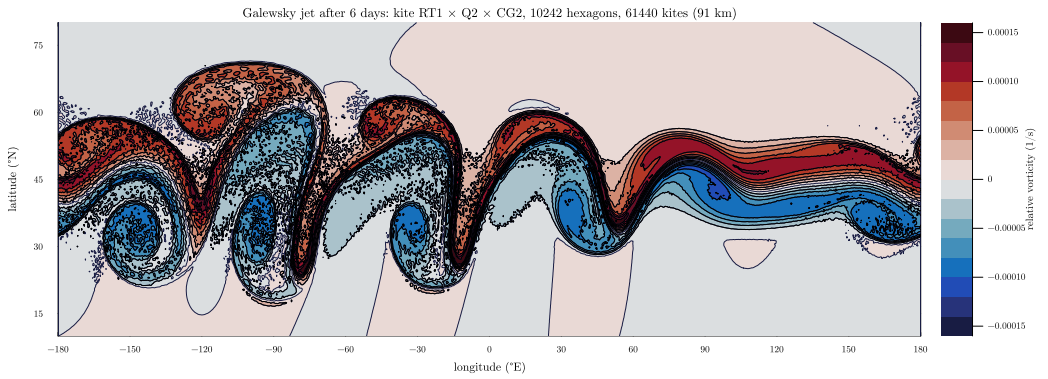


Figure 2: Galewsky jet, relative vorticity after 6 days, $\mathrm{RT}_1 \times Q_2 \times \mathrm{CG}_2$ on 10 242 hexagons (61 440 kites), no dissipation.

Comparison with the mixed method on the kite quadrilaterals. The kite mesh is an ordinary quadrilateral mesh, so the compatible pair $\mathrm{RT}_1 \times \mathrm{DG}_1 \times \mathrm{CG}_2$ with global mass matrices runs on it unchanged. On hexagon kites 2 and 3 it is about 1.9× more accurate ($1.94 \cdot 10^{-4}$, $4.72 \cdot 10^{-5}$) with fewer degrees of freedom, but needs a global sparse factorization; the kite method replaces it by independent dual-cell blocks, which matters most for explicit time stepping, on GPUs and in parallel.

Galewsky jet. The barotropically unstable mid-latitude jet of Galewsky et al. [6], without diffusion, on 10 242 hexagons (61 440 kites, about 91 km), SSP-RK3 with $\Delta t = 33.5\,\mathrm{s}$ (Figure 2). The jet breaks up after day 4 and rolls up into the familiar vortices by day 6. After 6 days the relative change of mass is $1.2 \cdot 10^{-11}$ and of energy $9 \cdot 10^{-10}$, from the time integration only. Without any dissipation, grid-scale noise in the vorticity accompanies the vortices.

Bickley jet. A doubly periodic $[0, 4\pi)^2$, $u = \operatorname{sech}^2(y - 2\pi)$, no rotation, Froude number 0.2, perturbed in wavenumber 1. On 64×64 quadrilaterals (16 384 kites) the jet forms a vortex street with two cat's-eye pairs by $t = 40$ (Figure 3); mass and energy change by $1.6 \cdot 10^{-10}$ and $7 \cdot 10^{-11}$.

7. The compressible Euler equations

7.1. Equations

The dry compressible Euler equations of a shallow atmosphere over flat ground, in height coordinates, with the horizontal velocity $\mathbf{u}$, the vertical

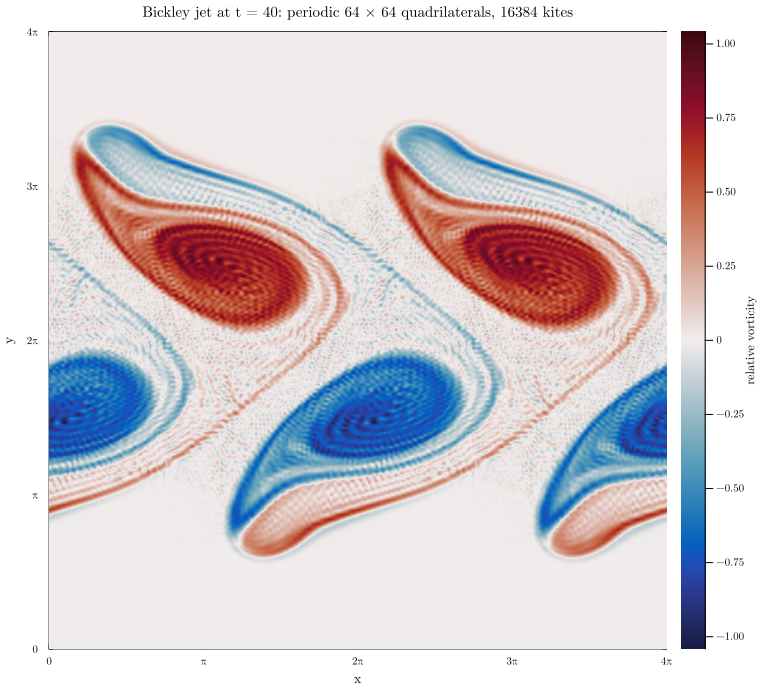


Figure 3: Bickley jet, relative vorticity at $t = 40$ on 64×64 quadrilaterals (16 384 kites), no dissipation.

velocity w, the density ρ, the potential temperature θ and the Exner pressure $\pi = (R\rho\theta/p_0)^{R/c_v}$, are

$$\partial_t \mathbf{u} = -(\zeta + f)\,\hat{\mathbf{r}} \times \mathbf{u} - w\,\partial_z \mathbf{u} - \nabla K - c_p \theta \nabla \pi, \tag{6}$$

$$\partial_t w = -\mathbf{u} \cdot \nabla w - w\,\partial_z w - c_p \theta\,\partial_z \pi - g, \tag{7}$$

$$\partial_t \rho = -\nabla \cdot (\rho \mathbf{u}) - \partial_z(\rho w), \qquad \partial_t(\rho\theta) = -\nabla \cdot (\rho\theta \mathbf{u}) - \partial_z(\rho\theta w), \tag{8}$$

with $K = \frac{1}{2}|\mathbf{u}|^2$, between the ground $z = 0$ and a rigid lid $z = z_T$, with $w = 0$ at both. They are the shallow-atmosphere equations with the traditional approximation: the distance from the centre is replaced by the radius a in all layers, $f = 2\Omega \sin\varphi$ is the only Coriolis term, and the metric terms u^2/r of the w equation and uw/r of the $\mathbf{u}$ equation are dropped. This is the consistent shallow system that conserves energy and angular momentum [20]; all 3D test cases below are steady states or perturbed steady states of exactly this system. A deep atmosphere, with $r = a + z$ in the metric and the full Coriolis vector, changes the extruded spaces only through the weights

r^2 in the vertical inner products and $1/r$ in the horizontal operators of each layer. They conserve the total energy $\int \rho(\frac{1}{2}|\mathbf{u}|^2 + \frac{1}{2}w^2 + c_v\theta\pi + gz)$, in which $c_v\rho\theta\pi$ is the internal energy and $\partial(c_v\rho\theta\pi)/\partial(\rho\theta) = c_p\pi$.

7.2. Extruded spaces

The vertical grid has the interfaces $0 = z_1 < z_2 < \cdots < z_{L+1} = z_T$ (index j), the layers (z_k, z_{k+1}) of thickness Δz_k with centres $\bar{z}_k$ (index k), and the interface thickness $\Delta z^h_j = \bar{z}_j - \bar{z}_{j-1} = (\Delta z_{j-1} + \Delta z_j)/2$. In the vertical, DG_0 is spanned by the indicator functions χ_k of the layers and CG_1 by the hat functions ϕ_j of the interfaces, with ϕ_1, ϕ_{L+1} removed for $w = 0$. The 3D spaces are

field	space	degrees of freedom
ρ, $\rho\theta$	$W_3 = H_p \otimes \mathrm{DG}_0$	Radau nodes × layers
w	$H_p \otimes \mathrm{CG}_1$	Radau nodes × interior interfaces
$\mathbf{u}$	$U_k \otimes \mathrm{DG}_0$	velocity degrees of freedom × layers

Since $\partial_z \mathrm{CG}_1 = \mathrm{DG}_0$, the 3D velocity space $W_2 = (U_k \otimes \mathrm{DG}_0) \oplus (H_p \otimes \mathrm{CG}_1)$ is mapped by the discrete divergence onto W_3, as in the compatible extruded elements [11, 10], with the horizontal part B and the vertical part ∂_z. θ lives with ρ in the layers (Lorenz staggering). The inner products are tensor products of the lumped rule (1) with the trapezoidal rule of every layer, the Gauss–Lobatto rule of CG_1. Both have their nodes at the degrees of freedom, so every Radau node carries one column of ρ, $\rho\theta$ and w that is coupled to the others only by the horizontal terms. This is the vertical counterpart of the mass lumping in the horizontal, and it is what a HEVI time step needs.

7.3. Horizontal terms

Per layer the horizontal terms are those of the shallow-water scheme (5) with ρ in place of h:

$$
\begin{aligned}
M_u\dot{\mathbf{u}} &= -\tfrac{1}{a}B^T K - \Big(\int \tfrac{\zeta+f}{\rho}\,(\hat{\mathbf{r}}\times\mathbf{F})\cdot\mathbf{v}_j\Big)_j - c_p\, M_u\, \Pi_U\Big(\theta\, M_u^{-1} B^T \pi/a\Big), \\
\mathbf{F} &= \Pi_U(\rho\mathbf{u}), \qquad M_h\dot{\rho} = B\mathbf{F}/a, \qquad M_h(\dot{\rho\theta}) = B\,\Pi_U(\theta\mathbf{F})/a.
\end{aligned}
\tag{9}
$$

With $E = \sum_k \Delta z_k \Big[\int \rho|\mathbf{u}|^2/2 + \sum_i M_{h,ii}(c_v\rho\theta\,\pi + \rho g\bar{z}_k)_i\Big]$ one has $\partial E/\partial\mathbf{u} = M_u\mathbf{F}$, $\partial E/\partial\rho = M_h(K + g\bar{z})$ and $\partial E/\partial(\rho\theta) = M_h c_p\pi$, and the horizontal terms exchange energy without loss, since the pressure pair cancels through

the symmetric $\Theta = (\int \theta\, \mathbf{v}_i \cdot \mathbf{v}_j)$. The vertical coupling $-\boldsymbol{\eta} w$ with the horizontal vorticity $\boldsymbol{\eta} = \partial_z \mathbf{u} - \nabla w$ and its counterpart $\boldsymbol{\eta} \cdot \mathbf{u}$ in the w equation are discretized as exact adjoints. Section 8 shows that (9) is not sufficient in three dimensions; Section 9 replaces the transport terms.

7.4. Vertical terms as a Galerkin method

Fix a Radau node i and drop the index; the horizontal mass $M_{h,ii}$ multiplies every term of the column and cancels. All vertical integrals use the trapezoidal rule of each layer.

Mass flux. The vertical mass flux is the lumped L^2 projection of ρw into CG_1, $F = \Pi_{\mathrm{CG}_1}(\rho w)$, the vertical counterpart of $\mathbf{F} = \Pi_U(\rho \mathbf{u})$. Testing with ϕ_j,

$$\Delta z_j^h F_j = \tfrac{1}{2}\Delta z_{j-1}\rho_{j-1} w_j + \tfrac{1}{2}\Delta z_j \rho_j w_j,$$

so that

$$F_j = \bar{\rho}_j w_j, \qquad \bar{\rho}_j = \frac{\Delta z_{j-1}\rho_{j-1} + \Delta z_j \rho_j}{\Delta z_{j-1} + \Delta z_j},$$

with the thickness-weighted interface mean $\bar{\rho}_j$. The flux of $\rho\theta$ is $G = \Pi_{\mathrm{CG}_1}(\theta F)$, $G_j = \bar{\theta}_j F_j$, with the same mean of θ.

Continuity. Testing $\partial_t \rho + \partial_z F = 0$ with χ_k is exact, since $\partial_z F \in \mathrm{DG}_0$:

$$\Delta z_k \dot{\rho}_k = -(F_{k+1} - F_k), \qquad \Delta z_k (\dot{\rho\theta})_k = -(G_{k+1} - G_k),$$

with $F_1 = F_{L+1} = 0$. In matrix form $M_3 \dot{\rho} = -D_z F$ with $M_3 = \mathrm{diag}(\Delta z_k)$ and $(D_z F)_k = F_{k+1} - F_k$.

Vertical momentum. The pressure gradient is the adjoint of the divergence, weighted with θ as in the horizontal. The discrete vertical gradient of $\pi \in \mathrm{DG}_0$ is $-M_2^{-1} D_z^T \pi$ with the lumped mass $M_2 = \mathrm{diag}(\Delta z_j^h)$, i.e. $(\pi_j - \pi_{j-1})/\Delta z_j^h$, and its product with θ is projected into CG_1 with the same rule:

$$\Delta z_j^h \dot{w}_j = -c_p \bar{\theta}_j (\pi_j - \pi_{j-1}) - g \Delta z_j^h - \Delta z_j^h r_j w_j,$$

where r_j is the absorbing layer of Klemp et al. [7] below the lid. These are the formulas of a finite-difference Lorenz grid, but every weight is fixed by the Galerkin method: the interface means are the projections of the trapezoidal rule, Δz_j^h is the lumped mass, and the pressure gradient is the adjoint of the divergence.

Theorem 3. *Without damping, the vertical terms exchange energy exactly: with $\partial E/\partial w = M_2 F$,*

$$\sum_j \Delta z_j^h F_j \dot{w}_j + \sum_k \Delta z_k \Big(c_p \pi_k (\dot{\rho\theta})_k + g \bar{z}_k \dot{\rho}_k \Big) = 0.$$

Proof. Summation by parts with $F_1 = F_{L+1} = 0$ gives $\sum_k c_p \pi_k \Delta z_k (\dot{\rho\theta})_k = -c_p \sum_k \pi_k (G_{k+1} - G_k) = c_p \sum_j \bar{\theta}_j F_j (\pi_j - \pi_{j-1})$, which cancels the pressure term of $\sum_j \Delta z_j^h F_j \dot{w}_j$, and $\sum_k g \bar{z}_k \Delta z_k \dot{\rho}_k = g \sum_j F_j (\bar{z}_j - \bar{z}_{j-1}) = g \sum_j \Delta z_j^h F_j$, which cancels the gravity term. □

The proof needs the same rule in the mass of w, in the flux F and in the means $\bar{\theta}$: the consistent CG_1 mass of w with the lumped flux, or an upwind θ in the flux with the mean $\bar{\theta}$ in the pressure gradient, break the identity. This is the vertical form of the quadrature condition of Section 8.1.

Upwinding. The central flux $\bar{\theta}_j F_j$ is the DG_0 flux with the mean trace. An upwind option adds the penalty $P_j = \alpha |F_j| (\theta_{j-1} - \theta_j)/2$, which for $\alpha = 1$ and equidistant layers turns $\bar{\theta}_j F_j + P_j$ into the upwind flux. The penalty does not change ρ and damps the variance of θ, at the cost of the exact energy exchange.

Tracers. A tracer ρq is transported with the same flux, $Q_j = \bar{q}_j F_j$ (plus the penalty), so that $q = 1$ reproduces the continuity equation.

7.5. Time discretization

The time step is the Wicker–Skamarock RK3 scheme [19] with the stages $\tau = \Delta t/3, \Delta t/2, \Delta t$, each a HEVI step from the state y^n with the tendencies of the stage state y^s:

1. explicit horizontal momentum, $\mathbf{u} = \mathbf{u}^n + \tau\, T_u(y^s)$;
2. forward–backward horizontal transport with the new velocity and ρ^s, θ^s: the explicit parts ρ^*, $(\rho\theta)^*$;
3. one implicit Galerkin step of the vertical terms of Section 7.4 in every column.

The implicit step freezes the interface means $\bar{\rho}_j$, $\bar{\theta}_j$ and the penalty P_j at the stage state, off-centres the vertical velocity with β, $w_j^\beta = \beta w_j + (1-\beta) w_j^n$,

and linearizes the Exner pressure around y^n, $\pi = \pi^n + \gamma^n(\rho\theta - (\rho\theta)^n)$ with $\gamma^n = (R/c_v)\pi^n/(\rho\theta)^n$:

$$\rho_k = \rho_k^* - \frac{\tau}{\Delta z_k}(F_{k+1} - F_k), \qquad (\rho\theta)_k = (\rho\theta)_k^* - \frac{\tau}{\Delta z_k}(G_{k+1} - G_k),$$
$$F_j = \bar{\rho}_j w_j^\beta, \qquad G_j = \bar{\theta}_j F_j + P_j,$$
$$w_j = w_j^n + \tau\Big(T_{w,j}(y^s) - g - r_j w_j\Big)$$
$$-\frac{\tau c_p \bar{\theta}_j}{\Delta z_j^h}\Big[\beta(\pi_j - \pi_{j-1}) + (1-\beta)(\pi_j^n - \pi_{j-1}^n)\Big].$$

Inserting $(\rho\theta)_k$ into the linearized π leaves one tridiagonal system in w per column. The flux F_j of the solved w is the one of the continuity, $\rho\theta$ and tracer equations, so mass is conserved to round-off and $q = 1$ stays exactly 1. The default is $\beta = 0.65$.

Discrete hydrostatic balance. The initial columns solve the discrete balance of the column step, $c_p\bar{\theta}_j(\pi_j - \pi_{j-1}) = -g\Delta z_j^h$, upwards from the lowest layer by a fixed-point iteration for π_j with the given temperature, so that an atmosphere at rest stays at rest to round-off.

8. From two to three dimensions

The 3D model with (9) conserves energy and keeps the rest state, but is unstable in a flow, although every shallow-water test passes. The 3D model is a tensor product, but four properties that are irrelevant or hidden in shallow water become essential:

1. the consistency of the quadrature of the velocity mass with the projections (Section 8.1);
2. the skew-symmetry of the projected transport of scalars without a gravity-wave feedback (Section 8.2);
3. the stability of the momentum transport in flow faster than the gravity waves (Section 8.3);
4. the different weighting of the sub-cell height modes by the horizontal and the vertical pressure gradient (Section 8.4).

Two further points are inherited from the vertical discretization: the Lorenz staggering of θ has a computational mode in the potential temperature [1, 14], and the vertical order is one below the horizontal one. The tool of the

analysis is the spectrum of the linearized horizontal tendency of one layer; $\max \operatorname{Re}\lambda > 0$ is a growing mode of the semi-discrete equations, independent of the time step.

8.1. Quadrature of the velocity mass

The velocity mass was first assembled with an exact rule (degree 6), the projections $\Pi_U(h\mathbf{v})$ with the 3×3 rule of the nonlinear terms. On non-affine kites the metric $J^TJ/\det J$ is rational, the two differ, and $\Pi_U(c\,\mathbf{v}) \neq c\,\mathbf{v}$ for a constant c ($1.4\cdot 10^{-2}$ on hexagon kites). Then the $\rho\theta$ flux $\Pi_U(\bar\theta\mathbf{F})$ is not $\bar\theta$ times the mass flux: the horizontal fluxes create sub-cell entropy at rest. The energy is still conserved, but the Casimir $\int \rho f(\theta)$ that makes the stratified rest state stable is not, and a sub-cell mode lives on the released potential energy. On the affine kites of the cubed sphere both rules are exact and the mode is absent. Assembling M_u with the rule of the projections restores $\Pi_U(c\,\mathbf{v}) = c\,\mathbf{v}$ to round-off and removes the mode. In shallow water there is no second scalar, and the defect is invisible.

With this rule the blocks of M_u are no longer the 2×2 node blocks of the lumped mass of [18] but the full dual-cell blocks of Table 1, also for the Q_p^2 space. They are sparse. A basis function couples only with the functions of its own kite, and two kites of a dual cell share only the s normal degrees of freedom of their common primal face ($s = k+1$ for RT_k and BDM_k, $s = p+1$ for Q_p^2). The graph of a block with n_w kites is therefore a ring of n_w cliques of $m = n/n_w + s$ unknowns that overlap in the face unknowns, with

$$\operatorname{nnz}(A) = n_w\left(\tfrac{m(m+1)}{2} - \tfrac{s(s+1)}{2}\right)$$

entries in the lower triangle, about $3/(2n_w)$ of the dense triangle. Numbered kite by kite, the Cholesky factor L gets fill only where the ring closes: s^2 entries for every kite beyond the third, so $\operatorname{nnz}(L) = \operatorname{nnz}(A) + s^2(n_w - 3)$. A dual cell of three kites (hexagonal meshes) has no fill at all, because every two faces of it belong to a common kite. A minimum-degree ordering (CHOLMOD) gives the same counts. Table 3 lists the blocks for the dual cells of the three meshes: the triangles and pentagons of the dual cells of icosahedral triangle meshes ($n_w = 6, 5$), the quadrilaterals of the cubed sphere ($n_w = 4$) and the triangles of hexagonal meshes ($n_w = 3$).

The sparse factor is computed once, because M_u does not change in time. A solve with it, one forward and one backward substitution per block and layer, reads $2\operatorname{nnz}(L)$ values instead of the n^2 of the dense inverse, 3 to 4.5

Table 3: Dual-cell blocks of the velocity mass with the rule of the projections, for n_w kites per dual cell: size n, nonzeros of the lower triangle of the block A and of its Cholesky factor L (kite-by-kite numbering; the fill is $s^2(n_w-3)$), of the dense lower triangle, multiply-adds of the factorization ($\sum_j c_j^2$ sparse, $n^3/3$ dense) and values read by a solve ($2\,\mathrm{nnz}(L)$ with the sparse factor, n^2 with the dense inverse). BDM_1 and Q_1^2 have the same counts.

space	n_w	n	$\mathrm{nnz}(A)$	$\mathrm{nnz}(L)$	dense	factorization	solve
BDM_1, Q_1^2 ($s=2$)	3	18	99	99	171	613 / 1944	198 / 324
	4	24	132	136	300	848 / 4608	272 / 576
	5	30	165	173	465	1083 / 9000	346 / 900
	6	36	198	210	666	1318 / 15552	420 / 1296
$U_1 = \mathrm{RT}_1$ ($s=2$)	3	30	225	225	465	1951 / 9000	450 / 900
	4	40	300	304	820	2632 / 21333	608 / 1600
	5	50	375	383	1275	3313 / 41667	766 / 2500
	6	60	450	462	1830	3994 / 72000	924 / 3600
Q_2^2 ($s=3$)	3	45	495	495	1035	6339 / 30375	990 / 2025
	4	60	660	669	1830	8551 / 72000	1338 / 3600
	5	75	825	843	2850	10763 / 140625	1686 / 5625
	6	90	990	1017	4095	12975 / 243000	2034 / 8100
$U_2 = \mathrm{RT}_2$ ($s=3$)	3	63	882	882	2016	14712 / 83349	1764 / 3969
	4	84	1176	1185	3570	19715 / 197568	2370 / 7056
	5	105	1470	1488	5565	24718 / 385875	2976 / 11025
	6	126	1764	1791	8001	29721 / 666792	3582 / 15876

times fewer for $n_w \geq 5$ and about 2 times for $n_w = 3$. The factorization itself costs $\sum_j c_j^2$ multiply-adds (c_j the entries of column j of L) instead of $n^3/3$, 3 to 22 times fewer, the more the larger the block. That matters where a block has to be factorized in every stage, as the density-weighted mass of Section 9. All blocks of one kind of dual cell have the same pattern up to the numbering of their unknowns: with the kites and their unknowns numbered in the same order in every dual cell, the blocks can be sorted by n_w and the symbolic factorization done once per kind, as in batched solvers for many small systems of one pattern [17]. Only the values of L are then stored per block, layers first, so that the threads of the layers of one block read the same pattern and adjacent values.

8.2. Projected transport of scalars

A passive tracer ρq transported like $\rho\theta$, $M_h(\dot{\rho q}) = B\,\Pi_U(q\mathbf{F})/a$, grows exactly like θ' in a one-layer flow, although its equation is linear and has no

feedback: the transport itself is unstable. The spectrum of the transport of a height field with a prescribed velocity (Williamson case 2, $\mathrm{RT}_1 \times Q_2$), in units of $u_0/\Delta x$:

$\max \mathrm{Re}\,\lambda$	hexagons	cubed sphere	triangles
projected flux $B\,\Pi_U(\rho\mathbf{u})$	0.700	0.278	1.473
pointwise flux, interior trace	4.757	1.245	2.428
DG, central trace	0.001	0.000	0.002
DG, upwind trace	0.000	0.000	0.000

The projected flux tests $\rho\mathbf{u}$ with the discrete gradient $M_u^{-1}B^T\eta$ of the test function, which carries the jumps of η across primal edges spread over the dual-cell mass blocks, so $\langle\dot\rho,\eta\rangle+\langle\dot\eta,\rho\rangle \neq \int \mathbf{u}\cdot\nabla(\rho\eta)=0$: a consistency defect at the grid scale, independent of the quadrature (degree 10 changes nothing) and largest on the irregular kites of triangles. In shallow water and in the 3D model the gravity-wave coupling to the momentum keeps ρ bounded; a transported quantity without that feedback (q, θ') sees the bare operator.

8.3. Supercritical flow

With the DG transport of θ (Section 9) the steady thermal wind still grows in 3D, on all three meshes and faster on triangles. Ablations show that the growth needs neither shear nor vertical advection nor rotation, but more than one layer: one layer in solid-body rotation stays at rest, five layers grow. The internal gravity waves of vertical mode m in an atmosphere of depth H and buoyancy frequency N_b travel at about $c_m = N_bH/(m\pi) \approx 105/m\,\mathrm{m/s}$ for $H = 30\,\mathrm{km}$, so the higher modes are slower than the jet. This is reproduced by one layer with a uniform flow $(1, 0.3)$ on the periodic unit square and the shallow-water equation of state with wave speed c: it is stable for $c > |\mathbf{u}|$ and grows for $c < |\mathbf{u}|$, at a rate roughly proportional to c. At $c = 0.5$:

$\max \mathrm{Re}\,\lambda\,\Delta x/\lvert\mathbf{u}\rvert$	full	momentum (ρ frozen)	continuity ($\mathbf{u}$ frozen)
quadrilaterals	3.49	0.77	0.77
triangles	6.08	5.48	5.49

Both transports of (9) are unstable on their own, the vector-invariant momentum transport (PV flux and ∇K) and the projected continuity $B\,\Pi_U(\rho\mathbf{u})$. In subcritical flow the fast gravity-wave coupling and the conserved energy

keep the coupled system neutral; this is why all shallow-water tests (Froude number below one) and the one-layer 3D tests pass. In supercritical flow the perturbations can draw energy from the mean flow. In the energy norm $E = \mathrm{diag}(HM_u, gM_h)$ the symmetric part of EJ is 35 % of its skew part, almost all of it in the momentum block; the continuity block contributes 0.3 %. Both blocks must be skew-symmetric on their own.

8.4. Sub-cell height modes

For $p = k + 1$ the horizontal pressure gradient $B^T\pi$ does not see the component of π along the cell modes ψ_P of Theorem 1, while the vertical gradient $(\pi_j - \pi_{j-1})/\Delta z_j^h$ acts on the nodal values and sees every pattern. In shallow water the modes are invariants and stay at the level of the initial projection. In three dimensions the vertical terms act node by node and change the coefficients $\psi_P^T M_h \rho$ and $\psi_P^T M_h(\rho\theta)$ of every layer, and a sub-cell pattern of π that is created by the vertical terms is not removed by any horizontal pressure force. The DG transport of Section 9 couples the modes to the flow, since its test functions are differentiated inside the kites with the pointwise velocity.

9. Discontinuous Galerkin transport

Scalars. ρ, $\rho\theta$ and the tracers are transported by

$$\langle \dot{(\rho q)}, \eta \rangle = \frac{1}{a}\Big(\sum_K \int_K q\, \mathbf{F}\cdot\nabla\eta - \sum_{F\in\mathcal{F}_p} \int_F \hat{q}\,(\mathbf{F}\cdot\mathbf{n}_K)\,[\![\eta]\!] \Big), \tag{10}$$

over the primal faces $\mathcal{F}_p$ with the central or upwind trace $\hat{q}$. For $q = 1$ it is $B\mathbf{F}$, so $\rho q = \rho$ is kept; it is skew in q (central) or dissipative (upwind). The pressure gradient uses its adjoint, so the energy exchange stays exact, and the horizontal advection of w is DG in advective form for the same reason. In the one-layer thermal shallow-water test (Williamson case 2, random $\theta' = 10^{-8}$) θ' now stays at 10^{-8} on all three meshes, instead of overflowing within a day on hexagons and triangles.

Momentum and density. With the DG momentum transport, ρ, $\rho\theta$ and the tracers are transported by (10) with the pointwise velocity $\mathbf{F} = \mathbf{u}$ and $\hat{q}$ the density trace, and the momentum equation is solved in advective form with

the density-weighted velocity mass:

$$\int \rho\,\dot{\mathbf{u}}\cdot\mathbf{v} = -\frac{1}{a}N(\mathbf{u};\mathbf{v}) - \int \rho f\,(\hat{\mathbf{r}}\times\mathbf{u})\cdot\mathbf{v} \\ - \frac{c_p}{a}\Big(\sum_K \int_K \rho\theta\,\mathbf{v}\cdot\nabla\pi - \sum_{F\in\mathcal{F}_p}\int_F \widehat{\rho\theta}\,(\mathbf{v}\cdot\mathbf{n}_K)[\![\pi]\!]\Big) \\ - \sum_{j=k,k+1}\frac{1}{2\Delta z_k}\int \rho\,w_j\,[\![\mathbf{u}]\!]_j\cdot\mathbf{v}, \quad (11)$$

$$N(\mathbf{u};\mathbf{v}) = \sum_K \int_K \rho\,\mathbf{v}\cdot(\mathbf{u}\cdot\nabla)\mathbf{u} - \sum_{F\in\mathcal{F}}\int_F \hat{\rho}\,\{\{\mathbf{u}\cdot\mathbf{n}\}\}\,[\![\mathbf{u}]\!]\cdot\{\{\mathbf{v}\}\} \\ + \alpha\sum_{F\in\mathcal{F}}\int_F \hat{\rho}\,\frac{|\mathbf{u}\cdot\mathbf{n}|}{2}\,[\![\mathbf{u}]\!]\cdot[\![\mathbf{v}]\!],$$

over all four faces $\mathcal{F}$ of every kite (across primal faces only the tangential component jumps, across dual faces the whole velocity), $\hat{\rho}$ the mean of the two traces and $\alpha = 1$ for upwinding. The last term of (11) is the vertical advection of the DG_0 velocity with the central flux: the jump $[\![\mathbf{u}]\!]_j = \mathbf{u}_j - \mathbf{u}_{j-1}$ between the layers $j-1$ and j is shared by the two layers. The pressure term is the adjoint of the DG $\rho\theta$ transport, so the exchange with the internal energy stays exact. The volume term is metric free except for the derivative of the Piola factor, $(\mathbf{u}\cdot\nabla)\mathbf{u}\,dA = \sum_k \hat{u}_k\,\partial_{\xi_k}(P\hat{\mathbf{u}})\,d\xi$ with $P = J/\det J$, $\partial_{\xi_k}(P\hat{\mathbf{u}}) = (\partial_{\xi_k}P)\hat{\mathbf{u}} + P\partial_{\xi_k}\hat{\mathbf{u}}$, and $\partial_{\xi_k}P$ from a Richardson extrapolation of central differences of the cell map. The density-weighted mass $\int \rho\,\mathbf{v}_a\cdot\mathbf{v}_b$ is assembled with the rule of the nonlinear terms and factorized per dual cell (Cholesky), so it stays local.

For uniform ρ and divergence-free $\mathbf{u}$ the central part of N is skew-symmetric, and the upwind penalty only removes energy. Since the kinetic energy of the advective form is not exactly the K of the pressure term (the projection of K into H_2), the scheme does not conserve energy exactly; the upwind term supplies the dissipation.

Figure 4 shows the spectra of the one-layer test at $c = 0.5$. The vector-invariant form has growing modes up to $3.49\,|\mathbf{u}|/\Delta x$ on quadrilaterals and $6.08\,|\mathbf{u}|/\Delta x$ on triangles. The DG form with the central trace is neutral on quadrilaterals and nearly so on triangles (0.014, from the quadrature on non-affine kites); with the upwind trace $\max \mathrm{Re}\,\lambda = 0$ on both. The DG advection has the explicit limit of DG with RK3, about 0.1 in the advective

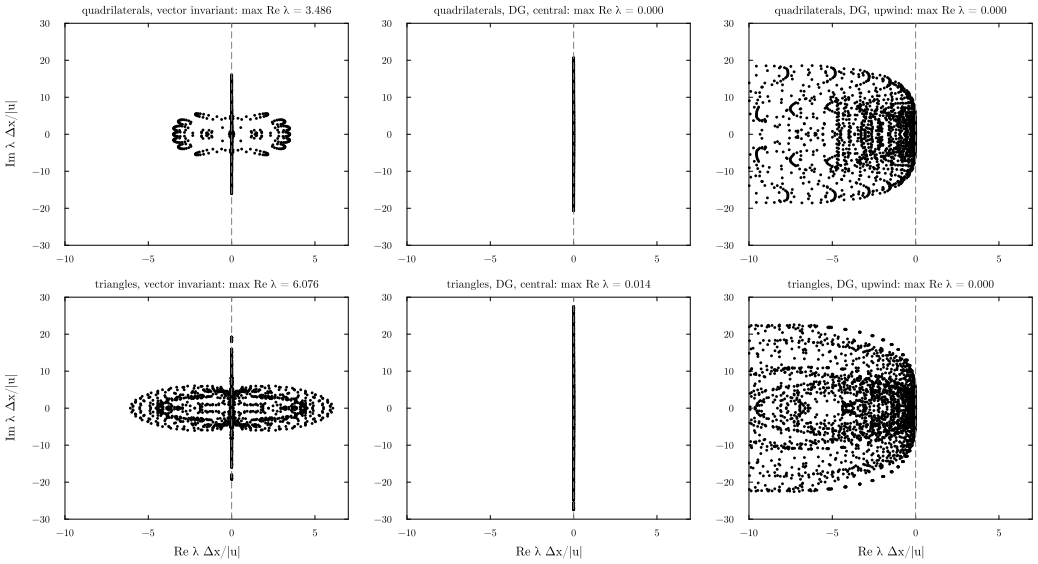


Figure 4: Spectra of the linearized one-layer tendency in a uniform supercritical flow ($|\mathbf{u}| = 1.04$, wave speed $c = 0.5$, periodic 6×6 mesh), in units of $|\mathbf{u}|/\Delta x$: vector-invariant form (9) and DG form (10)–(11) with central and upwind traces, on quadrilaterals (top) and triangles (bottom).

Courant number per kite; the 3D runs below are at about 0.02. It costs about 20 % more per step than (9).

10. Three-dimensional experiments

All runs use $\mathrm{RT}_1 \times Q_2$ horizontally, 30 equidistant layers up to 30 km, the absorbing layer above 22 km, the DG transport with upwind traces for θ, ρ, $\mathbf{u}$ and w, and no explicit diffusion. They are serial CPU runs.

10.1. Thermal wind

A steady rotating wind in exact gradient-wind balance, $1/T = (1 + \varepsilon \sin^2\varphi \operatorname{sech}^2(z/z_s))/T_0$ with $T_0 = 300\,\mathrm{K}$, $\varepsilon = 0.1$, $z_s = 12\,\mathrm{km}$ and the same surface pressure everywhere; the westerly reaches about 24 m/s. A good model keeps it at rest in the rotating frame (Figure 5). With the vector-invariant momentum, $\max|w|$ grows on all meshes, to $3 \cdot 10^{-2}$ (hexagons), 0.20 (cubed sphere) and NaN (triangles, day 2.5). With the DG momentum it stays at about 10^{-4} m/s on all three meshes, $\max|\theta'|$ at $2 \cdot 10^{-2}\,\mathrm{K}$ and $\max|\Delta p_s|$ at $5 \cdot 10^{-2}\,\mathrm{hPa}$.

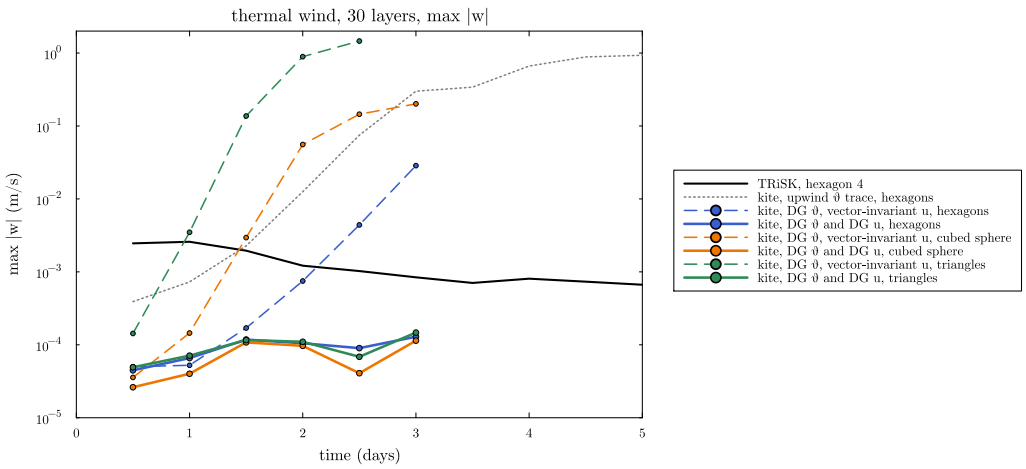


Figure 5: Thermal wind: max $|w|$ on hexagons 3 (23 682 columns), the cubed sphere 12 (21 600) and triangles 3 (24 320), with the upwind θ trace (grey), the DG θ transport with the vector-invariant momentum (dashed) and the DG transport of θ, ρ and $\mathbf{u}$ (solid).

10.2. Baroclinic wave in an f-plane channel

The channel of Ullrich, Reed and Jablonowski [16] is 40 000 km long and 6000 km wide, with a jet of 35 m/s, $f_0 = 1.0313 \cdot 10^{-4}\,\mathrm{s}^{-1}$, flat ground and a Gaussian perturbation of u (1 m/s, 600 km). The kite model has no walls, so we use a doubly periodic box of twice the width with the mirrored channel and the easterly jet; $v = 0$ at the walls then holds by symmetry, and the unperturbed state is an exact steady state. For this setup linear quasi-geostrophic theory gives the most unstable wave at zonal wavenumber 10 (4000 km) with the growth rate $6.59 \cdot 10^{-6}\,\mathrm{s}^{-1}$ [16]. Figure 6 shows the amplitude of this wave in the surface pressure on 60×18 cells (667 km), $\Delta t = 200$ s. After an adjustment of two days it grows at 5.9–$6.6 \cdot 10^{-6}\,\mathrm{s}^{-1}$, reaching $6.58 \cdot 10^{-6}$ at day 11, and saturates after day 12. The triangles follow the quadrilaterals to plotting accuracy; with the vector-invariant momentum the triangle run is dominated by a grid-scale mode from day 1. Figure 7 shows the wave in the surface pressure and the temperature at 500 m. It grows as a packet downstream of the perturbation, and from day 12 the warm air is drawn into tongues that roll up by day 14 with a minimum surface pressure of 955.6 hPa. On the triangles the minima are within 0.7 hPa (979.4 hPa at day 12, 955.8 hPa at day 14). From day 14 they show small ripples in the temperature at the warm edge of the front.

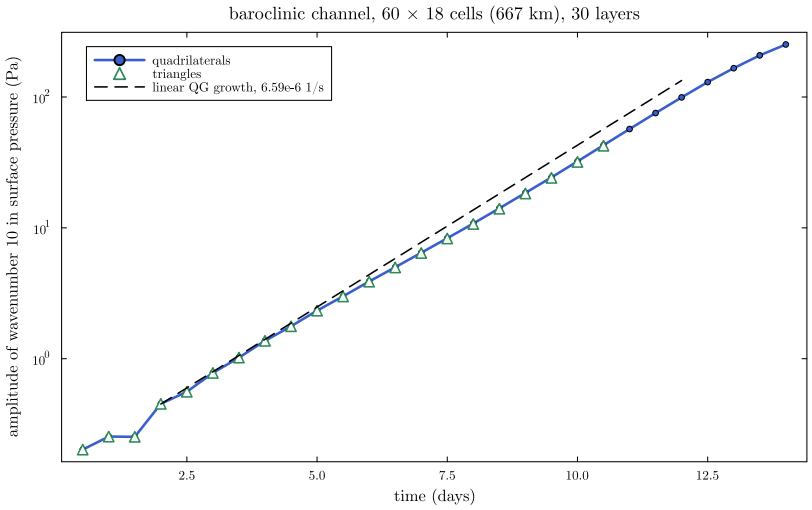


Figure 6: Baroclinic channel: amplitude of zonal wavenumber 10 of the surface pressure on 60×18 quadrilaterals (27 000 columns) and triangles (41 040 columns), and the growth rate of linear QG theory.

10.3. Baroclinic wave on the sphere

The baroclinic wave of Ullrich et al. [15] with the perturbation of DCMIP 2016, on hexagons 3, the cubed sphere 12 and triangles 3 (about 23 000 columns, roughly 130 km between height nodes), $\Delta t = 150$ s. Minimum surface pressure (hPa):

day	6	7	8	9	10
hexagons (23 682 columns)	993.5	984.3	968.3	941.6	925.8
cubed sphere (21 600)	993.4	984.1	966.9	942.7	926.3
triangles (24 320)	993.4	984.7	967.3	936.0	925.4

The three meshes agree to 1 hPa at day 10 (Figures 8 and 9); mass is conserved to 10^{-15}. The maps show the Q_2 fields evaluated inside the kites, not the nodal values. A lowest-order run (Q_1 heights) on hexagon 3 reached only 976.6 hPa, with $\max|w|$ near 1 m/s from grid-scale noise.

11. Related methods

Table 4 compares the kite method with the two closest ones. All three obtain local mass matrices from a staggered pair of macro-element meshes.

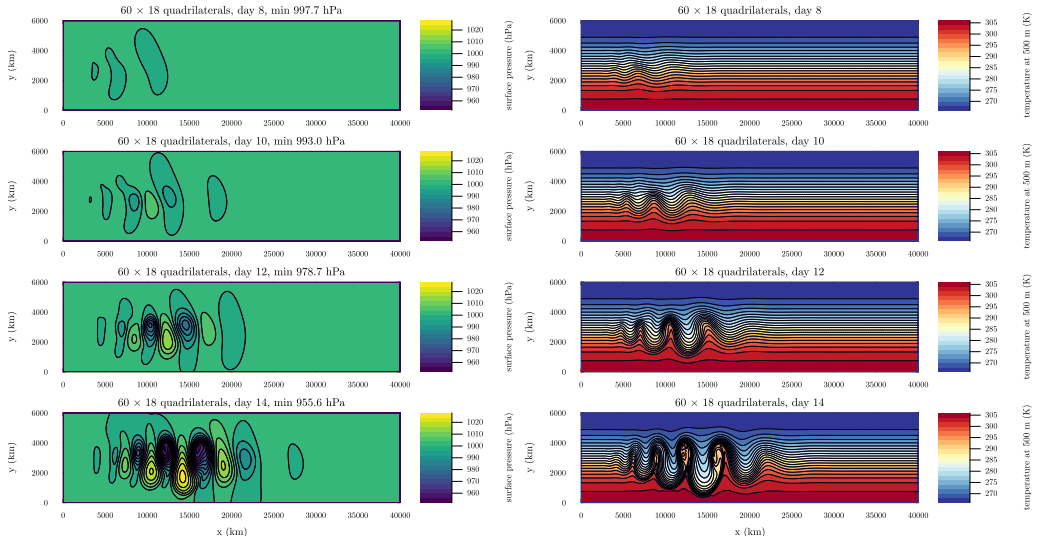


Figure 7: Baroclinic channel on 60×18 quadrilaterals: surface pressure (left) and temperature at 500 m (right) at days 8, 10, 12 and 14, the channel half $0 \le y \le 6000$ km of the doubly periodic box.

The equal-order pairs (the pair of [18], the $\mathrm{BDM}_1 \times P_1$ pair of [8]) have no spurious pressure modes; for the kite pair $\mathrm{RT}_k \times Q_{k+1}$ the velocity splits exactly into gradients and curls, and the price is the N stationary height modes of Theorem 1. The vertical part of the kite model is the lowest-order compatible extruded pair [11, 10], but with the trapezoidal rule in place of exact integration, so that the vertical mass is diagonal and the columns at the Radau nodes decouple; its formulas coincide with those of a Lorenz-staggered finite-volume column. The C-grid finite-volume models [13, 12, 22] have the same mass and energy properties at first order on general meshes.

12. Conclusions

The kite spaces give staggered finite elements on arbitrary polygonal meshes with diagonal height and block-diagonal velocity mass matrices. For $\mathrm{RT}_k \times Q_{k+1}$ the velocity space splits exactly into discrete gradients and curls, with one stationary height mode per primal cell. In shallow water the scheme conserves mass and energy exactly, converges at second order on hexagonal, triangular and quadrilateral meshes and runs on GPUs and with MPI. The vertical part, derived as a Galerkin method with the trapezoidal rule, keeps

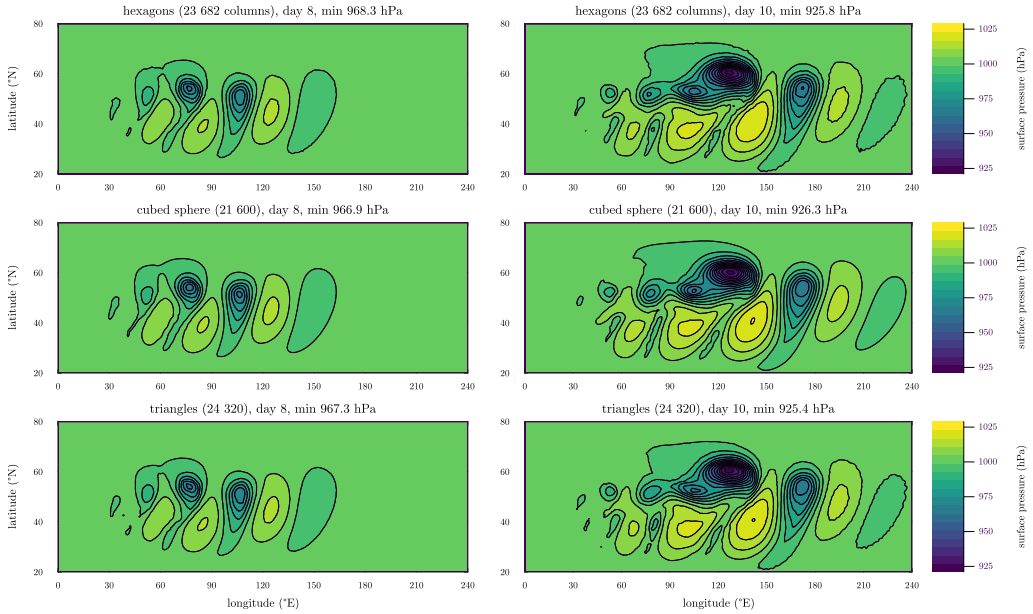


Figure 8: Baroclinic wave, surface pressure at days 8 and 10 on the three kite meshes (contour interval 6 hPa).

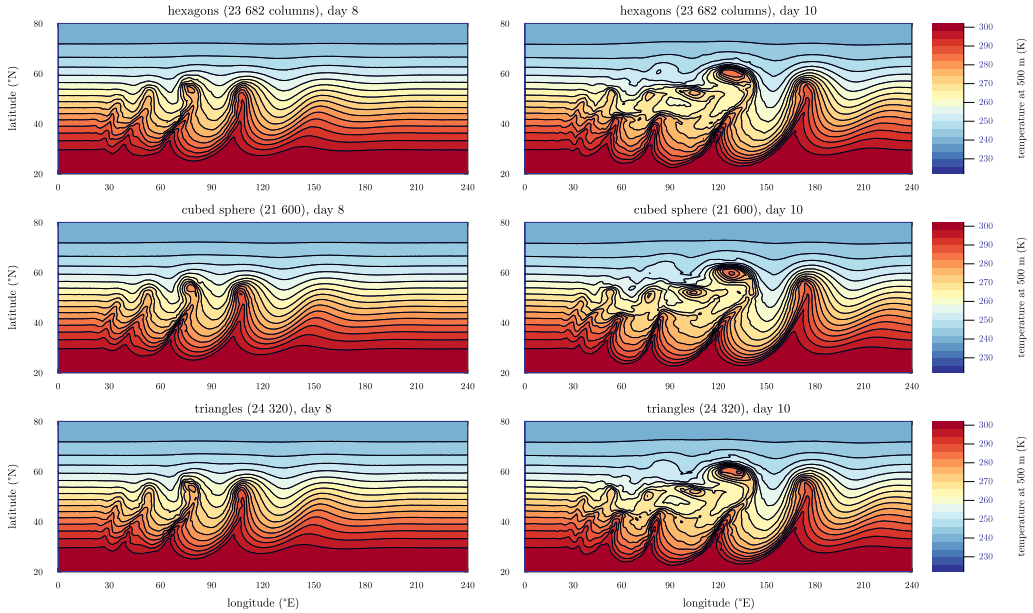


Figure 9: Baroclinic wave, temperature of the lowest layer (500 m) at days 8 and 10 on the three kite meshes (contour interval 4 K).

Table 4: The kite method and the two closest staggered methods.

	Wess et al. [18]	Leng, Lan, Ju [8]	kite method
mesh	triangles, barycentric dual cells around the vertices	polygons split into triangles around an interior point; dual elements: the two triangles of an edge	polygons split into kites; dual cells around the vertices, orthogonal dual where it exists
continuity	one field conforming on the primal mesh, the other in the dual cells	velocity normal-continuous in the polygons, height continuous in the edge pairs	height continuous in the polygons, velocity normal-continuous in the dual cells
spaces	lumped Q_p scalars and $Q_p \times Q_p$ vectors, any p	BDM_1 velocity, P_1 height	RT_k velocity, Q_{k+1} Radau heights (BDM and the pair of [18] as options)
mass matrices	diagonal / small blocks	block diagonal	diagonal height, dual-cell velocity blocks
geometry	plane	sphere, curved triangles	sphere, curved kites, metric-free Piola
equations	acoustics, Maxwell	nonlinear shallow water	nonlinear shallow water, compressible Euler

the column structure of HEVI schemes and exchanges energy exactly. The horizontal transport must be built with care in three dimensions: the velocity mass must use the rule of the projections, scalars must be transported by DG, and in supercritical flow, which the higher vertical modes of a stratified atmosphere always are, the energy-conserving vector-invariant momentum transport is unstable. A DG transport of momentum and density with the pointwise velocity and upwind traces removes this instability on all meshes. With it the model is quiet on the thermal wind, reproduces the linear baroclinic growth rate in a channel and gives the same baroclinic wave on three different meshes.

Open points: the DG momentum form is not energy conserving, because the kinetic energy of the advective form differs from the projected K of the pressure term; whether an energy-conserving skew-symmetric form with the same stability exists is open. The sub-cell height modes are seen by the vertical but not by the horizontal pressure gradient; a pair without them, or a filter of the modes, is to be tested in 3D. The DG terms are serial CPU

loops so far. Charney–Phillips staggering, with θ in the space of w, and higher vertical order are the next steps for the vertical part.

Code availability

`dmplex.jl` (`https://github.com/OsKnoth/dmplex`), the models `NonlinearKiteShallowWater` and `KiteNonhydrostatic`. The examples `galewsky.jl`, `bickley.jl`, `thermal_wind.jl` and `baroclinic_channel.jl` run the test cases; the figure scripts are in `docs/paper/scripts`.

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